

Nyx Haile

AI Systems & Research Engineer

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EDUCATION

Massachusetts Institute of Technology

2026

B.S. Computer Science, B.S. Mathematics, Minor in Economics

Cambridge, MA

- Coursework: algorithms, theory of computation, computer systems security, cryptography, statistics.

EXPERIENCE

Partial

2025 – Present

Cofounder & CEO

London

- Co-founded Partial, a prediction market that turns any relational analysis between assets into a tradable position.
- Built the Rust pricing engine, which prices two-asset trades over ~200k-cell probability grids in 50-150 ms on a 2-CPU server.

Weiss Laboratory & The SHED, MIT

June 2025 – Sept 2025

Undergraduate Researcher (UROP), generative AI for vascularised liver tissue

Cambridge, MA

- Built the fluidics-design AI agent and wrote the master prompt for a proof-of-concept multi-agent LLM workflow that generates complex vascular STL designs from plain-text specifications; co-inventor on the MIT invention disclosure.
- Wrote a parametric vascular-network generator specification for the Cellvivo bioreactor, covering stochastic branching, tapering, R-tree collision avoidance and a 0.99 diffusion-coverage loop.

Vana

Dec 2022 – July 2023

AI Developer

Remote

- Shipped an iMessage sticker app in Swift on the Vana API, generating personalised stickers from users' AI portraits.
- Engineered and iterated prompt templates for personalised AI-generated sticker portraits, tuning style and expression cues for identity retention and consistency.

MITERS, MIT Radio Society & SIPB

Sept 2021 – Present

Keyholder (MITERS, Radio Society); SIPB member

Cambridge, MA

- Contributed to arkOS, SIPB's self-hosted AI assistant, introducing a typed StateOutput model that its agent state machine returns and consumes.

SELECTED PROJECTS

Repositories marked *private* are available on request.

Strand (*private*) | Rust, FoundationDB

Capability-authenticated issue tracker

- Built a federable, capability-authenticated issue tracker in Rust on FoundationDB, with line-level blame and content-addressed rationale.
- Shipped a 16-crate, ~71k-line Rust workspace through v1.0.0 GA (10/10 verification gates) and v1.1.0, then a v2.0 governed-execution prerelease.

Forge (*private*) | Rust, systemd, RPC

Local-first agent control plane

- Built a Rust daemon, operator CLI, Claude Code hooks and provider proxy that coordinate coding-agent work across multiple git repositories.
- Implemented an atomic file-write tool that records attribution for every write and reverts the file if recording fails; the default recorder fails closed.

howtowin.lol | Python, PyTorch, CUDA, Redis

World model for post-game teaching

- Designed a Dreamer-lineage RSSM world model for League of Legends combining static context, player priors and event sequences with outcome, next-event, next-frame and next-decision heads.
- Trained the world model end to end on 12.8k games; at minute 15 it predicted the winner at 0.88 AUC on unseen matches and 0.84 on unseen players.

Sherlock | Rust, Dolt

Token-usage forensics for agent CLIs

- Built a Rust CLI that ingests agent-CLI logs (Claude Code, Codex, Copilot), attributes token usage per session, and reports what burns tokens without adding value.
- Stored ingested usage in a Dolt versioned-SQL repository.

TECHNICAL SKILLS

Languages: Rust, Python, TypeScript, SQL **ML:** PyTorch, NumPy, CUDA, GPU training and retrieval **Systems:** Linux/systemd, Docker, FoundationDB, PostgreSQL, SQLite, Redis **Web:** SvelteKit, React, Next.js